Examiner only

11. *(a)* Esters are formed when carboxylic acids are heated with alcohols in the presence of concentrated sulfuric acid. This is a reversible reaction that is carried out by heating the reagents under reflux.

(i) Draw a labelled diagram of the apparatus you would use to carry out a reaction under reflux. [3]

(ii) Explain how the apparatus that you have drawn in part (i) results in the process of reflux. [1]

....

....

(iii) Why is it necessary to use a reflux technique in this type of reaction? [1]

....

....

(iv) State the function of the concentrated sulfuric acid in this reaction. [1]

....

(b) The equation shows the reaction between ethanoic acid and methanol.

$$CH_3COOH + CH_3OH \rightleftharpoons CH_3COOCH_3 + H_2O$$

When 3.00 g of ethanoic acid was heated under reflux with 1.28 g of methanol in the presence of concentrated sulfuric acid for 10 minutes it was found that 1.18 g of methyl ethanoate was formed.

(i) Suggest how the methyl ethanoate could be separated from the mixture formed in the reaction vessel. [1]

....

(ii) Calculate the percentage of theoretical yield of methyl ethanoate obtained. Show your working. [3]

Percentage yield = ... %

(iii) Suggest **one** change that could be made to this preparation to improve the percentage yield of methyl ethanoate. Explain your suggestion. [2]

....

....

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(c) Two reactions of organic compound **A** are shown below.

$$\mathbf{B} \xleftarrow{\text{conc } H_2SO_4} \mathbf{A} \xrightarrow{Cr_2O_7^{2-}/H^+} \mathbf{C}$$

Compound **B** is a straight-chain hydrocarbon with the formula C_4H_8.

(i) What type of reaction is taking place when compound **A** is heated with concentrated sulfuric acid to form compound **B**? [1]

....

(ii) Draw the displayed formulae of **two** possible isomers of **A**. [2]

(iii) What colour **change** is seen when compound **A** reacts with the $Cr_2O_7^{2-}/H^+$ solution? [1]

....

(iv) What type of reaction is taking place when compound **A** is heated with $Cr_2O_7^{2-}/H^+$ to form compound **C**? [1]

....

(v) Draw a displayed formula of compound **C**. [1]

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